

The Science Behind Every Decision You've Ever Regretted

A near-fall on a 14,000-foot cliff led me to real research on why teen brains are wired for risk.

THE PULSE · AUGUST 9, 2026 · BY ARPEET SHAH

We were coming down from the summit of Pikes Peak, at a 14,000 feet altitude. It's the kind of view that makes you forget how far the drop actually is. My brother stopped to pose for a picture near the edge. He's older, steadier, built for that kind of balance. He nailed it without thinking twice.

Nobody was taking a picture of me. But naturally, like a teen that tries to look cool, I tried the same pose anyway.

My foot slipped before I even registered what was happening. For one full second, I was two feet away from a 14,000-foot fall with absolutely nothing between me and the edge. I don't remember deciding to move. My body just threw itself sideways, and I hit the ground hard enough to leave bruises down my arm and across my face.

There is only one honest answer as to why I still have the opportunity to write this article. I got lucky.

But here's what I keep coming back to: nobody made me try that pose. There was no camera, no audience, no reason. I just wanted to see if I could do what my brother did. Some part of my brain thought that was worth the risk, standing that close to a 14,000-foot drop.

Have you ever sat down and thought about why teens love to take risks so much, even when deep down we know the reward is barely worth it? Parents have spent years trying to lecture us. But if our brains are just built this way, is it even fixable? Turns out there's a real reason my brain made that call in half a second. Surprisingly, it has nothing to do with being reckless.

Here's the thing that nobody tells you: your brain isn't one single thing making a decision, it's more like two systems running at the same time, and they don't always agree with each other.

One of these systems is like an alarm bell for excitement. Without taking in the factor of harm, your brain is just addicted to a risk that might not even benefit you after 10 minutes just because it "sounds cool." It's the reason a risky idea can feel genuinely irresistible in that moment.

The other system is more like your brain's "brakes." It's supposed to be the part of the brain that stops the first system from totally controlling your mind. It weighs consequences, imagines what could go wrong, and tries to stop you before you do something you'll regret.

In a fully grown adult brain, these two systems are roughly balanced, hence the drop in risk-taking. By the time the excitement settles in, the brakes are ready to be applied. But here's the fundamental problem: the two systems don't finish developing at the same time.

The excitement system, technically called the limbic system, matures early. By the time you're a teenager, it's not just working, it's running hotter than it will at any other point in your life. The braking system, called the prefrontal cortex, takes much longer. It doesn't finish developing until your mid twenties. So when you match the timelines up, there is a set of years, the teen years, where the limbic system is at full volume, and the prefrontal cortex is still under construction.

Now this isn't just a theory, a researcher decided to actually measure it. In 2010, a researcher named Laurence Steinberg ran a study with 935 people between the ages of 10 and 30. He tracked two things in both of them: how much they truly craved reward and excitement, and how much self control they really had.

What he found lines up exactly with the two systems that we just talked about. The craving for reward doesn't rise steadily and lay flat after that. It climbs during your childhood, peaks during your teen years, and then dies down as you become an adult. If you were to graph it, it would look like a hill.

Self control on the other hand, doesn't peak at all. It just climbs steadily starting from age 10 and keeps improving in a straight line, all the way to adulthood.

Put those two lines on the same graph, and one age jumps out immediately. Right around fifteen or sixteen, you get the worst possible combination, reward-seeking at its highest point, while self control still has years left to catch up. Steinberg's own conclusion was that heightened risk-taking in the middle of the teen years may come from exactly that combination, a brain that wants more than it ever will again, paired with a braking system that isn't finished yet.

So why does this gap exist? Why didn't evolution build both systems on the same timeline? There are two real answers and scientists genuinely disagree on which one is correct.

The first answer is that the gap is essentially an accident. The prefrontal cortex is one of the most complex parts of the human brain, so naturally it takes longer to develop. The gap between the two systems isn't there for a reason, it just happens when one part is more complex than another and therefore takes more time to evolve.

On the other hand, the second answer is a little more unique as it honestly is a little unsettling. Some scientists argue that the gap isn't an accident at all. It's on purpose.

Here is the logic. At some point, every human has to leave the safety of their family and survive by themselves. An adult-cautious brain would never do any of that. It would just stay at home, where it's safe. So maybe evolution turned up the reward system on purpose, specifically during the teen years, to push people out into the world right when they need to start building that independence. Under this theory, the risk-taking isn't a glitch, it's the whole point.

Neither side is truly settled yet, but the second theory makes us realize something important. It means that part of me that wanted to pose on the edge of that cliff may not be a flaw at all. It might be your brain doing exactly the job it evolved to do.

Here's where I've landed, at least for now. I don't think this is fully settled science, and I don't think it should be. But if I had to guess, I think the truth leans closer to something built on purpose. The evolutionary logic is what actually got me. Every human eventually has to leave the safety of home, and a fully cautious brain would never choose to do that. Somebody had to want to take that first step out, and it makes sense that biology would be the thing pushing for it.

The gap between these two systems is real either way. What's still up for debate is whether evolution meant for it to be there.

What I do know is this. Standing on that mountain, some part of my brain decided a pose was worth a 14,000-foot risk. At the time, it felt like a stupid decision. Now, I'm not so sure it was stupid at all. It might have just been my brain doing exactly what teenage brains are built to do, whether that was on purpose or not.

So the next time you do something a little reckless, a little impulsive, a little “why did I just do that,” maybe don't beat yourself up over it. Your brain isn't broken. It might just be exactly fifteen or sixteen years old, doing precisely what it's built to do.

Just maybe don't test it two feet from a 14,000-foot drop.

REFERENCES

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Originally published on The Pulse — arpeetshah.substack.com. Read more at arpeetshah.github.io/the-pulse/